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INSECURE ATTACHMENT AND DEPRESSION: SUPPORT FOR MEDIATIONAL ROLE OF SELF-ESTEEM

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Abstract – Insecure attachments to childhood caregivers are associated with later mental health risks, such as depression, but research is less clear on the mechanisms for the relationship between attachment and depression. One potential pathway by which attachment could relate to depression risk is through their respective relationships with self-esteem. The purpose of this study is to examine whether self-esteem mediates the relationship between attachment style and depression symptoms in a sample of college students. Participants included 138 undergraduate students from a small southern university. They completed an online survey that assessed attachment style dimensions, depression, and self-esteem. Results of this study showed a positive relationship between insecure attachment and depression symptoms, with self-esteem partially mediating that relationship. These findings suggest that insecure attachments may increase the risk for depression through an impact on self-esteem. This study adds a theoretical rationale to the body of work that indicates that improving relationship patterns and self-esteem contributes to the prevention and treatment of depression.

The National Health and Nutrition Examination Survey (2009-2012) estimates that approximately 7.6% of individuals aged 12 and over have depression (Pratt & Brody, 2014), but recent studies suggest that rates of mental health problems in college students are climbing. From 2013 to 2021, rates of positive screens for depression from the National Healthy Minds Study (HMS, 2021) increased from 17.4% in 2013 to 40.8% in the 2020-2021 academic year (Lipson et al., 2022). To stem the rising tide of mental health problems in young adults, it is crucial to identify mechanisms that contribute to depression that can be prevented or altered. One such pathway to the development of depression may come from attachment styles developed early in life through experiences with caregivers.

Attachment theory posits that early experiences with caregivers affect later relationships with significant others (Bowlby, 1980). Early in development, caregivers help individuals develop an internal working model of themselves and others that may influence self-esteem and future relationships (Riggs & Han, 2009). The internal working model is a framework through which individuals evaluate their worthiness, others' trustworthiness, and the goodness of the world via memories and experiences established through relationships with caregivers (Ainsworth et al., 1978). Receiving responsive care is likely to foster secure attachment as well as a positive internal working model (Riggs & Han, 2009).

On the other hand, those who receive inconsistent care or are rejected by their caregivers may develop insecure attachments and negative internal working models, characterized either by attachment anxiety or attachment avoidance (Bowlby, 1980). Attachment anxiety refers to the extent to which one is worried about being unloved or abandoned, and attachment avoidance refers to the extent to which one is comfortable with emotional closeness and interdependence (Zhu et al., 2016).

Insecure attachment styles may affect later relationships. Those with more attachment avoidance may be less affected by the feedback of others, including positive feedback. Those with higher attachment anxiety, however, may be more affected by negative opinions and feedback of attachment figures. Insensitivity to positive feedback and/or high sensitivity to negative feedback could contribute to a higher risk of depression and/or low self-esteem. Consistent with this theory, Hepper and Carnelley (2012) had adults keep a diary for 14 days and found that those with high anxious attachment scores were more affected by negative interpersonal feedback with romantic partners compared to adults low in anxious attachment, whereas those with high avoidant attachment were less impacted by positive interpersonal

feedback compared to those low in avoidant attachment. Insecure attachments may predispose individuals to interpret social feedback in a way that maintains low self-esteem and depression.

Insecure attachment styles are associated with greater depressive symptoms. Those with attachment anxiety believe that they are unworthy and are concerned that their attachment figure will abandon them, whereas those with avoidant attachment believe that others are not trustworthy and close relationships are unimportant (Riggs & Han, 2009). These negative cognitions about oneself and others could be contributing factors in the development of depression, particularly during adolescence when individuals are self-critical and aware of others' appraisals of them. Indeed, research has found that insecure attachment styles are associated with depression, and more secure attachments are linked to fewer depressive symptoms (e.g., Dagan et al., 2018; Erozkan, 2011). Erozkan's (2011) study specifically focused on college students and found that insecure attachment style was positively associated with depression and loneliness. Dagan et al. (2018) published a meta-analysis of 55 studies that assessed the relationship between adult attachment styles and depressive symptoms and concluded that there is a relatively consistent positive relationship between insecure attachment and depression.

Although the link between insecure attachments and depression has been well-established, less is known about the mechanisms by which insecure attachment styles are tied to depression risks in young adults. However, in adolescents, poorer relationships with parents are linked to higher depression (e.g., Hu & Ai, 2016; Kamkar et al., 2012), and these relationships were mediated by self-esteem. Similarly, self-esteem emerged as a mediator between parental conflict and depression in children (Ying et al., 2018). Identifying intervening mechanisms in the relationship between attachment and depression would provide more targets for intervention.

Previous research has identified variables that mediate insecure attachment and depression to be an increased need for reassurance from others and a lower capacity for self-reinforcement (Wei et al., 2005), dysfunctional attitudes about one's self (Lee & Hankin, 2009), and lower perceived social support (Zhu et al., 2016). These mediating variables have been linked to low self-esteem in other studies. For example, Clerkin et al. (2013) found greater need for reassurance on Facebook to be linked to lower self-esteem, and Kocouski and Endler (2000) found lower frequency of self-reinforcement to be linked to lower self-esteem in undergraduate students. Low self-esteem has also been linked to dysfunctional attitudes, such as maladaptive perfectionism (Ashby & Rice, 2002), and lower perceived social support (Ikiz & Cakar, 2010). Low self-esteem may be a common feature of mediators that have thus far have been linked to insecure attachment and depression.

Low self-esteem is both a symptom of depression (American Psychological Association, 2013) as well as a risk factor for depression (Beck, 1976). This is what Orth and Robins (2013) call the reciprocal relation model; low self-esteem makes one vulnerable to depression, and depression further erodes self-esteem. Low self-esteem was found to be a vulnerability factor for later depression in a longitudinal study of Mexican youth, even when controlling for extraneous variables (Orth et al., 2013). The negative view of self that is associated with insecure attachment (Roberts et al., 1996) could increase vulnerability to depression through lowered self-esteem. Those with secure attachments to caregivers likely have supportive caregivers who help them foster a positive self-concept. Insecurely attached individuals do not have this constant support from caregivers to buffer life stress and reinforce a positive self-concept. The way that people feel about themselves may be influenced by internal working models about the value of oneself that developed in the context of the relationship with the primary caregiver. Consistent with this theory, perceptions of a positive parental attachment are linked to higher self-esteem in adults (Foster et al., 2007), as well as fewer depressive symptoms (Kenny & Sirin, 2006).

With low self-esteem being both a risk factor for depression and a possible result of insecure attachment, it is likely to be a mediating factor between adult attachment and depression. To the best of our knowledge, there are no recent large-scale studies of self-esteem in college students, but a study published by Sprecher et al. (2013) that included a sample of 7,552 college students from 1990 to 2012 found that students generally had a relatively high level of self-esteem (mean score of 3.94 out of 5). Sprecher et al. reported that self-esteem remained consistent across the years of the study, but some other researchers have found that self-esteem rose between 1998 and 2008 (see meta-analysis by Gentile et al., 2010). Although self-esteem is relatively high in college students, at least up until 2012, low self-esteem is associated with problems, such as an increased vulnerability to depression and anxiety (Sowislo & Orth, 2013).

Studies are unclear whether interventions can change attachment style, but research has shown that self-esteem is amenable to change. Although a person's sense of security can be changed in the short-term in laboratory experiments (e.g., Mikulincer & Shaver, 2007),

longer-term attachment style changes seem to happen in the context of long-term romantic relationships (Arriaga et al., 2018). Self-esteem, however, is more directly changeable through targeted interventions. Niveau et al. (2021) published a meta-analysis that examined 119 published studies involving interventions to increase self-esteem. They found that interventions are generally effective across formats (online versus in-person, group versus individual, and shorter versus longer interventions). Larger effect sizes were found for cognitive-behavioral interventions compared to other types of interventions, and improvements to self-esteem after an intervention were generally maintained for up to one year after the intervention was completed.

The goal of this study is to determine whether there is support for a mediating role of self-esteem in the relationships between insecure attachment dimensions and depressive symptoms. This study aims to answer the following questions: (a) Are higher scores on measures of insecure attachment (anxious and avoidant) linked to lower self-esteem and greater depression symptoms? (b) Does self-esteem mediate the relationship between anxious attachment and depression? (c) Does self-esteem mediate the relationship between avoidant attachment and depression? We hypothesize that all research questions will be answered in the affirmative.

Method

Participants and Procedure

The sample consisted of 138 college students who completed measures of attachment style, self-esteem, and depression symptoms through the online survey website Qualtrics. Participants were aged 18 to 47 and collected from a convenience sample at a small, southern university. Informed consent was obtained prior to each person's participation. Participants were not directly compensated, though some may have been given extra credit in a course for participation. Participants completed all surveys in one 10–15-minute session at their own convenience. A demographics questionnaire was presented first to all participants, but the other measures were presented by Qualtrics in a randomized order. The sample consisted of 87% women, 11.6% men, and 1.4% who selected a response option listed as "other" or chose not to answer. The average age of the participants was 20 years old (SD = 3.6). A majority identified as white/Caucasian (52.9%; n = 73), followed by black/African American (32.6%; n = 45) and the response option "other"/biracial (13.7%; n = 19).

Measures

Adult Attachment Scale (AAS)

The 18-item AAS (Collins & Read, 1990) was used to assess participants' feelings about relationships. The measure presented statements to assess how comfortable the participant is with being close to others and depending on others as well as how much the participant worries about abandonment. Participants were asked to rate the extent to which the statement described them on a Likert-type scale from 1 (not at all characteristic) to 7 (extremely characteristic). Examples of statements included "I find it relatively easy to get close to others," "I find that people are never there when you need them," and "I want to merge completely with another person." In the current study, responses were scored using a scoring method developed by the authors of the measure consisting of two subscales, anxiety and avoid. The anxiety subscale measured the extent to which one worries about abandonment or unrequited love. The avoid scale measured the extent to which one is comfortable with intimacy and depending on others to be there when needed. Item scores were summed (after reverse-scoring certain items) and then divided by the number of items in the scale to yield an item average. Higher scores represent more insecure attachments (avoidant and anxious). The AAS was found to have an internal consistency greater than .58 for the subscales and a test-retest reliability of .70 over a period of two months (Ravitz et al., 2010). In our sample, the internal consistency of the AAS anxiety subscale was acceptable (α = .73) and the internal consistency of the avoid subscale was good (α = .84).

Beck Depression Inventory (BDI)

The BDI (Beck et al., 1961) consists of 21 self-report items and was administered to assess depressive symptoms. Participants were instructed to indicate the statement that best described their thoughts and feelings in the past month. Responses were numbered with higher numbers indicating greater severity of symptoms (e.g., 0 = I do not feel like a failure to 3 = I feel I am a complete failure as a person). Scores were summed to indicate the severity of depressive symptoms that may be present. Higher scores represented greater endorsement of depression symptoms. Internal consistency of the BDI has been found to be good, with a Cronbach's alpha of .86 for psychiatric populations and .81 for non-psychiatric populations (Beck et al., 1988). BDI total scores can be used to detect depressive symptoms in clinical patients as well as nonclinical populations, with clinical patients having higher scores (Kim & Park, 2010). The internal consistency of BDI scores was high in our sample (α = .92).

Rosenberg Self Esteem Scale (RSE)

The 10-item RSE (Rosenberg, 1965) was used to assess self-esteem. The measure presented statements regarding self-worth and self-image, and participants were asked to indicate the extent to which they agree or disagree with a statement on a Likert-type range from 1 (strongly disagree) to 4 (strongly agree). Sample items include "On the whole, I am satisfied with myself" and "At times I think I am no good at all." After reverse-scoring appropriate items, item scores were summed. Higher scores indicated higher self-esteem. Robins et al. (2001) found the RSE to have high internal consistency (α = .88) and to be stable through a 4-year period (r = .69). Internal consistency of RSE scores in our sample was high (α = .91).

Results

Descriptive Statistics

First, we calculated descriptive statistics for scores on all measures to understand how participants scored on the measures of depression, self-esteem, avoidance attachment, and anxious attachment. Participants' average scores on the avoid and anxiety attachment subscales were similar (2.82 and 2.75, respectively). Item scores can range from 1 to 5 with higher scores indicating more insecure attachment. The item means in our sample show that attachment tended to be slightly more secure than insecure, but there was sufficient variability in scores to be able to test our hypotheses. Self-esteem was fairly high with a mean score of 28 out of a possible range of 10 to 40. Participants reported a relatively low level of depression. Using Beck et al.'s (1988) recommended cut-off scores on the BDI, 46% reported no or minimal depression (scores of 0 to 9). However, 24.8% reported mild depression (scores of 10 to 18), 22.6% reported moderate depression (scores of 19 to 29), and 6.6% reported severe depression (scores of 31 to 63). Table 1 summarizes descriptive statistics on the BDI, RSE, and AAS.

Next, we calculated correlations between our variables to ensure that our variables related to each other in the expected directions based on previous research. This also served as a check to ensure that we had scored all measures correctly. All measures were significantly correlated with each other in the expected directions. Higher scores on measures of insecure attachment dimensions were associated with lower self-esteem and greater depressive symptoms. Table 2 summarizes correlations for BDI scores, RSE scores, anxious attachment, and avoidant attachment.

Table 1
Summary of Descriptive Statistics

	Mean	*SD*	Range
Anxious Attachment Item Average	2.75	0.87	1 – 5
Avoidant Attachment Item Average	2.82	0.70	1 – 5
Self-Esteem Total Score	28.38	6.14	13 – 40

Table 2
Correlations among Variables

	Depression	Self-Esteem	Anxious Attachment
Self-Esteem	-.72		
Anxious Attachment	.39	-.43	
Avoidant Attachment	.41	-.42	.30

Note. All results were statistically significant at a level of $p < .001$.

Table 3
Indirect Effects of Attachment on Depression through Self-Esteem

Attachment Dimension	Point Estimate	Standard Error	Lower CI	Upper CI
Anxious	.59	.11	.38	.83
Avoidant	.35	.07	.22	.52

Note. Results are statistically significant.

Self-Esteem as a Mediator between Insecure Attachment Dimensions and Depression

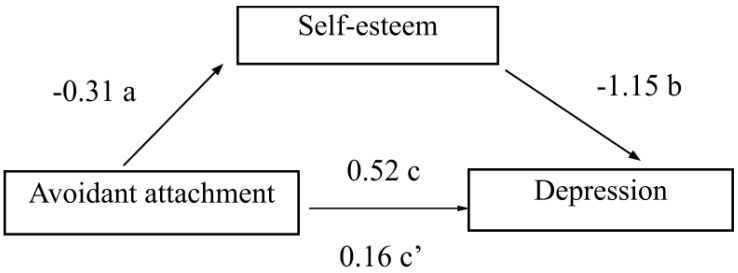
To address our research questions, we conducted mediation analyses to examine whether the relationships between insecure attachment dimensions and depression could be partially explained by self-esteem. Mediation analyses were conducted using the SPSS v.22 software with the PROCESS v2.16 macro for mediation (Hayes, 2013) with 5000 random bootstrap samples to address our hypothesis that self-esteem would mediate the relationship between adult insecure attachment styles and depression. A mediation analysis was conducted for each insecure attachment dimension (anxious and avoidant).

A bias-corrected 95% bootstrap confidence interval (CI) was calculated (See Table 3) for each

mediation analysis, and statistical significance was indicated when the CIs did not include zero. There was a significant indirect effect from avoidant attachment to depression symptoms through self-esteem ($\kappa^2 = .30$) and a significant indirect effect from anxious attachment to depression symptoms through self-esteem ($\kappa^2 = .31$).

Figure 1

Mediation Model of the Indirect Effect of Avoidant Attachment Scores on Depression Scores through Self-esteem Scores



Note. Path a demonstrates that greater avoidant attachment scores predict lower self-esteem, and path b shows that greater self-esteem corresponds to lower depression symptoms. There was a direct effect of attachment on depression scores (path c), but self-esteem was a significant partial mediator between attachment and depression. Self-esteem accounted for about one-third of the direct effect between avoidant attachment scores and depression scores (path c'). All paths were significant at $p < .01$.

Results supported our hypothesis of a significant indirect path from avoidant attachment scores to depression symptoms through self-esteem. As depicted in Figure 1, the overall model was significant, $F(2, 134) = 77.34$, $p < .001$, $R^2 = .54$. Higher avoidant attachment scores predicted lower self-esteem scores, and lower self-esteem scores were associated with higher depression scores. There was still a direct effect of avoidant attachment on depression scores when self-esteem was included in the model, but the indirect effect of avoidant attachment on depression through self-esteem was also significant. These results indicate support for a model of partial mediation of self-esteem in the relationship between avoidant attachment and depression.

Results also supported our hypothesis of a significant indirect path from anxious attachment to depression symptoms. As illustrated in Figure 2, the overall model was significant, $F(2, 133) = 75.18$, $p < .001$, $R^2 = .53$. Higher anxious attachment scores predicted lower self-esteem scores, and lower self-esteem scores were associated with higher depression scores. There was still a direct effect of anxious attachment on depression scores when self-esteem was included in the model, but the indirect effect of anxious attachment on depression through self-esteem was also significant. These results indicate support for a model of partial mediation of self-esteem in the relationship between anxious attachment and depression.

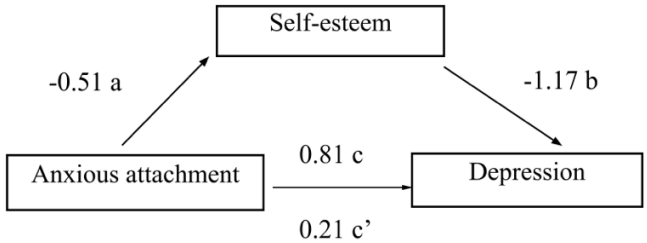
Discussion

This study examined the relationships among adult insecure attachment dimensions, self-esteem, and depressive symptoms. Results supported the hypothesis that more insecure attachment (anxious and avoidant) is associated with lower self-esteem and greater depression symptoms. We also found evidence to support the hypothesis that self-esteem partially explains the relationship between anxious and avoidant attachment styles and depression, so targeting self-esteem may be impactful in the treatment and prevention of depression.

Our descriptive data for self-esteem, anxious and avoidance attachment dimensions, and depression are consistent with scores in other non-clinical samples. The relatively high self-esteem scores in our sample are consistent with Sprecher et al.'s

Figure 2

Mediation Model of the Indirect Effect of Anxious Attachment Scores on Depression Scores through Self-esteem Scores



Note. Path a demonstrates that greater anxious attachment scores predict lower self-esteem, and path b shows that greater self-esteem corresponds to lower depression symptoms. There was a direct effect of attachment on depression scores (path c), but self-esteem was a significant partial mediator between attachment and depression. Self-esteem accounted for about one-fourth of the direct effect between anxious attachment scores and depression scores (path c'). All paths were significant at $p < .01$.

(2013) analysis of self-esteem in college students over a 22-year period. Similarly, the mean anxious and avoidant attachment item ratings that were mid-way between the anchors representing the highest and lowest ratings are similar to average item ratings in other studies (e.g., Collins & Read, 1990). Although the mean score for depression in our sample was relatively low (12.97 out of a possible high score of 63), over half of the sample (54%) screened positive for at least mild depression, with 29.2% of the sample screening positive for moderate to severe depression. This is consistent with other studies of college students conducted prior to Covid-19. A meta-analysis by Ibrahim et al. (2013) found reported prevalence rates of depression to be 10 to 85% in studies with college students, with a weighted mean prevalence of 30.6%. It is important to note that data collection for our study occurred prior to the Covid-19 pandemic. Numerous studies indicate that mental health problems rose across all segments of the population during the Covid-19 pandemic (e.g. Czeisler et al., 2020). It would be beneficial to repeat this study to determine if the effect sizes for relationships among insecure attachment dimensions, self-esteem, and depression would be even higher than what we found in our data that were collected prior to the Covid-19 pandemic.

The findings of this study are consistent with previous research that finds greater depressive symptoms and lower self-esteem to be associated with higher scores on measures of insecure attachment (e.g., Erozkan, 2011; Orth & Robins, 2013). Like other studies that assessed possible mediators of the relationship between insecure attachment and depression (e.g., Lee & Hankin, 2009; Roberts et al., 1996), attachment insecurity in our study was linked to more depressive symptoms partially through an association with other variables. In our study, low self-esteem was a partial mediator of the relationship between insecure attachment dimensions and depression symptoms. Low self-esteem was also found to have a direct link with depressive symptoms, consistent with past studies (Kenny & Sirin, 2006; Orth & Robins, 2013; Roberts et al., 1996).

The main findings of this study that support a mediational role of self-esteem between anxious and avoidant attachment and depression are important because self-esteem is amenable to change following short-term interventions (Niveau et al., 2021), whereas changes to attachment style take longer and seem to happen in the context of long-term romantic relationships (Arriaga et al., 2018). To our knowledge, there are no studies that demonstrate that attachment style can be changed by interventions that target low self-esteem, but the research on the general stability of attachment style over time (Zang & Labouvie-Vief, 2010) would suggest that it is unlikely that a short-term intervention to improve self-esteem would also change a person's attachment style. There is ample research, however, that demonstrates that improved self-esteem reduces the risk of depression (e.g., Orth et al., 2013). Furthermore, interventions to increase self-esteem usually decrease depressive symptoms at the same time (see systematic review and meta-analysis by Kolubinski et al., 2018). Identifying changeable mediating factors in the relationship between more intractable variables and mental health outcomes is important in efforts to reduce depression. Although attachment style does not change easily, knowing that insecure attachment styles may increase depressive symptoms through lowered self-esteem gives us an avenue for prevention and treatment of depression in people with insecure attachments.

This study is limited by use of self-report measures, few male participants, and its cross-sectional sample Though low self-esteem is linked to higher depression, our cross-sectional sample cannot determine that low self-esteem contributed to depression symptoms as opposed to depression depleting self-esteem. It is also difficult to determine if insecure attachment contributes to depression or if depression causes attachment problems in adults. Our findings, however, are consistent with studies that suggest that low self-esteem predicts later depression (e.g., Kenny & Sirin, 2006; Lee & Hankin, 2009). Additionally, self-esteem was found to be stable over a 4-year period (Rosenberg, 1965), supporting theories that self-esteem is a stable construct and not simply a product of depression. Our support for a mediational role of self-esteem between dimensions of insecure attachment and depression in our cross-sectional sample is bolstered by the stability of self-esteem and its ability to predict later depression. This study uses a non-clinical sample with a good range of scores on the BDI and RSE, and the predominantly female sample reflects the reality that women are about two times more likely to be diagnosed with major depressive disorder compared to men (Li et al., 2022). This sample is appropriate for examining factors that may contribute to depression and resilience.

Future studies should examine additional mediating factors to better understand how insecure adult attachment relates to depression symptoms. In our sample, self-esteem only accounted for a third or less of the relationship between attachment and depression symptoms. Adding perceived social support and traumatic life events as additional mediators could further explain the relationship between attachment and depression, as they can activate attachment behavior

(Riggs & Han, 2009) and affect one's utilization of supportive resources (Zhu et al., 2016), thereby affecting depressive symptoms. Future research should utilize longitudinal studies to establish more definitively whether self-esteem mediates the relationship between insecure attachment and later depression symptoms.

It is important to identify possible factors contributing to the development of depression and the mechanisms through which they operate. Insecure attachment is a risk factor for depression symptoms, and this study suggests that low self-esteem may be an intervening variable. Improving self-esteem in individuals with insecure attachments may be a salient target for preventing depression.

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